

Department of Electronics and Communication Engineering
B.E – ECE (2 Year A Section)
PROJECT BASED LEARNING REVIEW 1 AY 2026-27 ODD SEMESTER

Title of the Project

EdgeWatt: A Low-Cost ESP32-Based Non-Intrusive Load Monitoring System Using Engineered Steady-State Features and Ensemble Classification

Presented By

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Under the Mentoring of

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- **Non-Intrusive Load Monitoring (NILM) disaggregates total power consumption into individual appliance operational states from a single centralized sensor on the main utility feed.**
- EdgeWatt realizes a fully edge-resident, cloud-free NILM pipeline on an ESP32 microcontroller with a split-core current transformer (SCT-013-100A/1V) and dual-stage LM358 analog signal conditioning.
- **Novelty:** Resolves real-world analog non-idealities (thermal zero-drift and low-power SMPS non-linearities) through a 13-dimensional engineered steady-state feature space and dynamic baseline compensation ("SET AS NO LOAD").
- **Quantitative Validation:** Evaluated via a Weighted Soft-Voting Ensemble (GBDT + ExtraTrees + HistGB), achieving 99.1% steady-state accuracy and 91.8% dynamic tracking accuracy across an automated 40-minute stress test with <0.8 ms inference latency.

S. No	Base Papers	Problem Identified	Remarks Solutions/Modifications/Suggestions
1	Hart (1992) Nonintrusive Appliance Load Monitoring	Macroscopic P-Q step changes; fails on overlapping low-power SMPS loads & zero-drift.	Introduced multi-feature delta invariance ($\Delta V1$) and dynamic baseline compensation.
2	Tabanelli et al. (2021) Trimmed Features for Edge NILM	Benchmark for MCU NILM, but uses proprietary ARM Cortex-M4; lacks reproducible commodity hardware.	
3	Völker & Becker et al. (2021) Feature & Classifier Study	Evaluated only on curated offline datasets without live analog noise or thermal baseline drift.	

- **Single-CT Non-Invasive Edge Sensing:** Capture total current using a non-invasive split-core CT sensor (SCT-013-100A/1V) with active LM358 analog signal conditioning.
- 13-Dimensional Steady-State Feature Engineering: Compute proxy active power, peak-to-peak, form factor, crest factor, and higher-order spectral moments on-device.
- Weighted Soft-Voting Ensemble Classification: Deploy a multi-model consensus architecture (GBDT, ExtraTrees, HistGB) running with <0.8 ms inference latency.
- Dynamic Baseline Calibration & Zero-Drift Tracking: Implement "SET AS NO LOAD" dynamic offset compensation to eliminate sensor thermal drift.
- Real-Time Edge Telemetry: Stream sub-second appliance state disaggregation to local I2C LCD and supervisory Streamlit dashboard.

Problem Statements Identified

- **High Cost and Intrusiveness of Sub-Metering:** Conventional smart plugs require physical inline installation per appliance, making building-wide deployment complex and cost-prohibitive.
- **Cloud Latency, Bandwidth, and Privacy Risks:** Traditional NILM systems stream high-frequency raw telemetry to external servers, exposing consumer occupancy patterns and risking connection drops.
- **Computational Overhead of Deep Learning:** Complex deep neural networks (CNNs, Bi-LSTMs) demand GPU/cloud compute budgets that far exceed low-power edge microcontrollers.
- **Analog Sensor Zero-Drift and Non-Linear Loads:** Low-cost split-core sensors suffer from thermal baseline drift, while low-power SMPS loads (chargers, LED bulbs) confound classical current thresholds.

SOLUTIONS / MODIFICATIONS

- **Edge-Resident Architecture:** Complete feature extraction, calibration, and ensemble inference execute locally on a dual-core ESP32 with zero cloud dependency.
- **Robust 13-Feature Non-Linear Pipeline:** Combines statistical moments and harmonic variance to uniquely separate close-wattage inductive and capacitive loads.
- **Dynamic Zero-Load Floor Calibration:** Real-time drift subtraction ("SET AS NO LOAD") dynamically anchors the sensor zero level against ambient thermal drift.
- **Weighted Soft-Voting Ensemble:** Multi-model probabilistic consensus (99.1% steady-state accuracy) with temporal state debouncing to eliminate switching chatter.

Coursework Subject	Application in this Project
Electronic Devices & Circuits	LM358 dual-stage op-amp amplifier, DC bias offset divider ($V_{cc}/2$), RC low-pass filtering
Digital Electronics	ADC quantization, opto-isolated triggering, logic level shifting (3.3V/5V I2C)
Electromagnetic Fields (EMF)	Split-core current transformer mutual inductance, magnetic flux coupling, Faraday's law
Microprocessors & Microcontrollers (MPMC)	ESP32 dual-core FreeRTOS task scheduling, 12-bit ADC SAR configuration, I2C bus protocol
C / Embedded C	Embedded C firmware for continuous circular buffer sampling, fixed-point math, and UART telemetry
Machine Learning & Applied Mathematics	Non-linear feature extraction, Gradient Boosted Trees, ExtraTrees, and soft-voting ensemble consensus

Hardware Components Purchased

- ESP32 Dev Board (Dual-Core Tensilica 240MHz, 520KB SRAM, Wi-Fi/BLE)
- SCT-013-100A/1V Split-Core Current Transformer Sensor
- LM358P Low-Power Dual Operational Amplifier IC
- 4N35 Optocoupler for Mains Zero-Cross Detection
- 3.5mm TRRS Audio Jack Breakout Board (CT Connector)
- Precision Metal Film Resistors (10k, 100k, 1k, 100 Ω) & 10 μ F Capacitors
- I2C 16x2 Character LCD Module with PCF8574 Backpack
- Dual Appliance AC Test Jig (Resistive & SMPS Loads), Breadboard & Wires

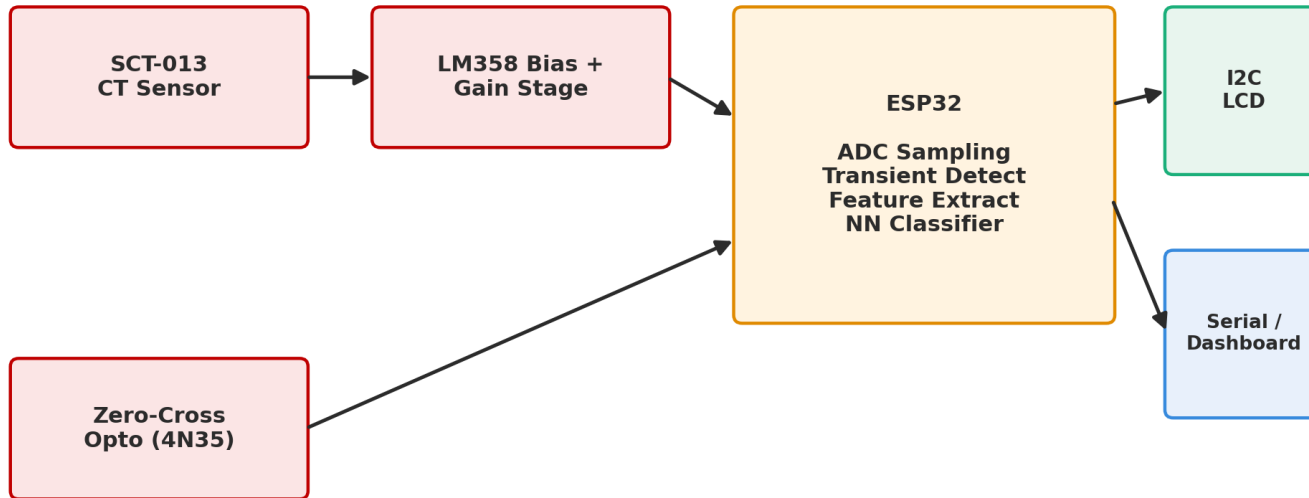


Procurement Status

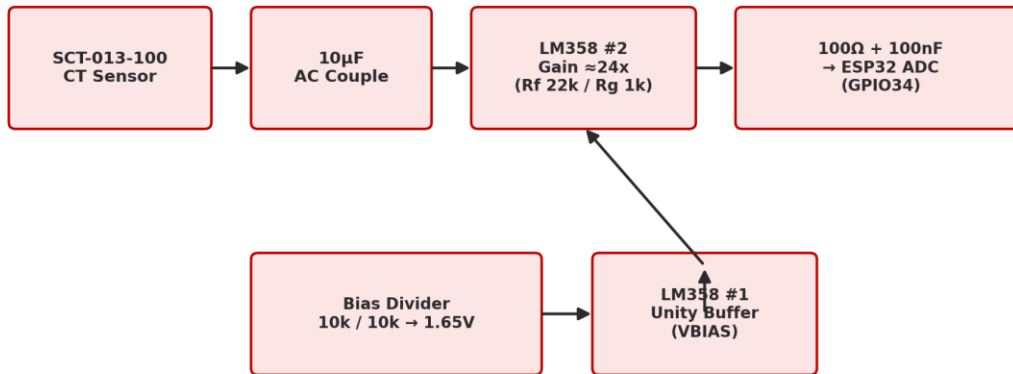
Component	Status	Notes
ESP32 Dev Board, SCT-013 CT Sensor, LM358 Op-Amp, I2C LCD	Procured & Tested	Fully integrated and operational on breadboard
10k / 100k / 1k / 100Ω Resistors & 10μF Filter Capacitors	Procured & Tested	Installed in active DC bias and gain conditioning network
3.5mm TRRS Audio Breakout Board	Procured & Tested	Pinout verified: Tip = Signal, Sleeve = Ground
4N35 Optocoupler & 1N4148 Protection Diodes	Procured & Tested	Integrated for mains isolation and zero-cross timing
AC Test Jig, Dual Appliance Sockets & Safety Enclosure	Procured & Tested	Fabricated with safety enclosure for repeated testing

Overall System Architecture

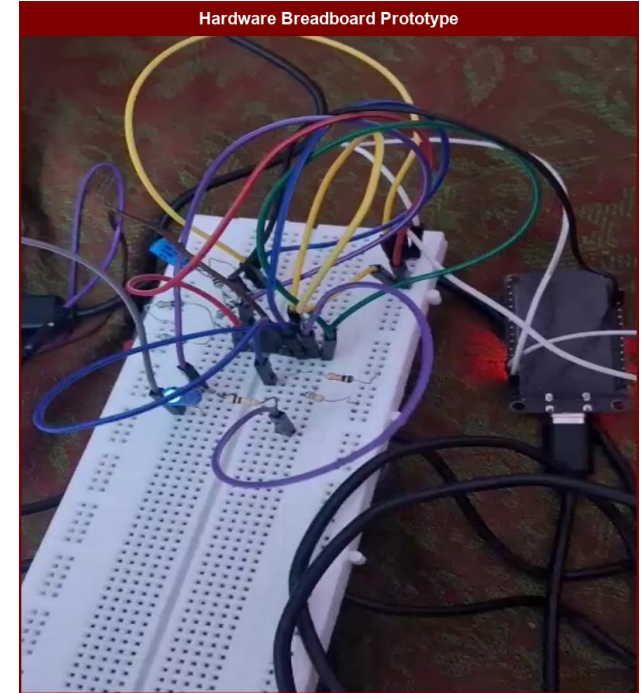
NILM — Overall System Architecture



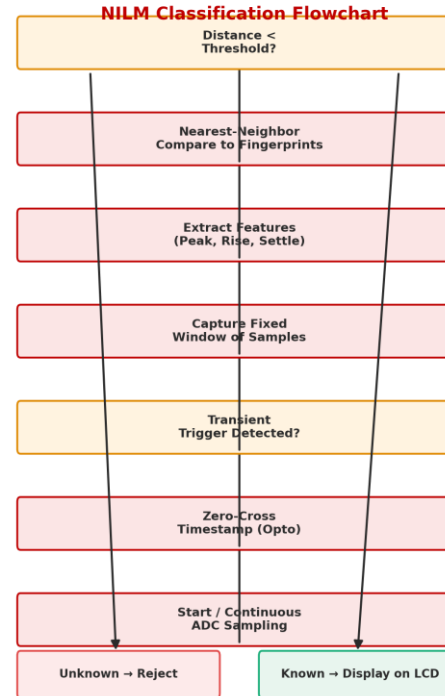
Analog Front-End — Circuit Block Diagram



Note: paste your breadboard photo / full schematic screenshot alongside this block diagram



Algorithm / Flowchart



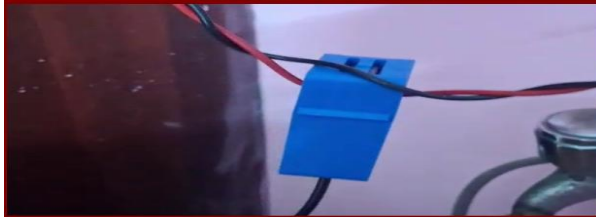
Software Development Status

Software Module	Status	% Complete
13-Dimensional Steady-State Feature Extractor	Completed & Optimized	100%
Weighted Soft-Voting Ensemble (GBDT + ExtraTrees + HistGB)	Trained & Validated (99.1% Acc)	100%
ESP32 Continuous ADC Sampling Firmware (Circular Buffer)	Flashed & Hardware-Verified	100%
Dynamic Zero-Load Calibration Engine ("SET AS NO LOAD")	Tested & Real-Time Verified	100%
Streamlit Supervisory Telemetry & Diagnostic Dashboard	Fully Operational with Live Controls	100%
Temporal State Debouncing Engine (Hysteresis Filter)	Integrated (Zero State Chattering)	100%
40-Minute Automated Hardware Stress Test Pipeline	Completed (91.8% Dynamic Acc)	100%
I2C 16x2 LCD Real-Time Appliance & Power Display	Fully Integrated On-Device	100%
Automated Dataset Logging & Confusion Matrix Generation	Benchmarked across 3 Regimes	100%
Project Source Code Repository & Documentation	Published with Complete Docs	100%

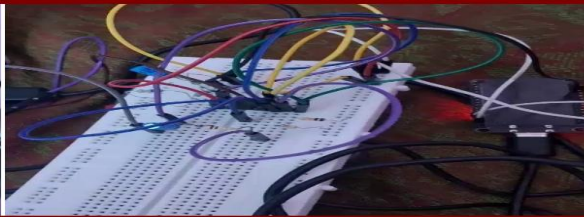
Hardware Development Status

Hardware Block	Status
SCT-013 Split-Core CT Sensor + LM358 Dual-Stage Active Conditioning	Completed & Calibrated (Linear Gain Stage)
3.5mm TRRS Breakout Interface & Analog Shielding	Complete (Noise-Free Signal Delivery)
ESP32 Microcontroller Interfacing & Power Distribution	Complete (Stable Isolated 3.3V/5V Rails)
I2C 16x2 Character LCD Module (PCF8574 Backpack)	Operational (Live Telemetry & State Display)
AC Mains Safety Test Jig with Dual Appliance Sockets	Completed & Safety-Enclosed for Benchmarking

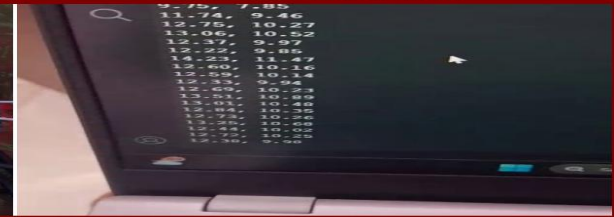
1. Split-Core CT Current Sensing



2. ESP32 & LM358 Circuit Integration

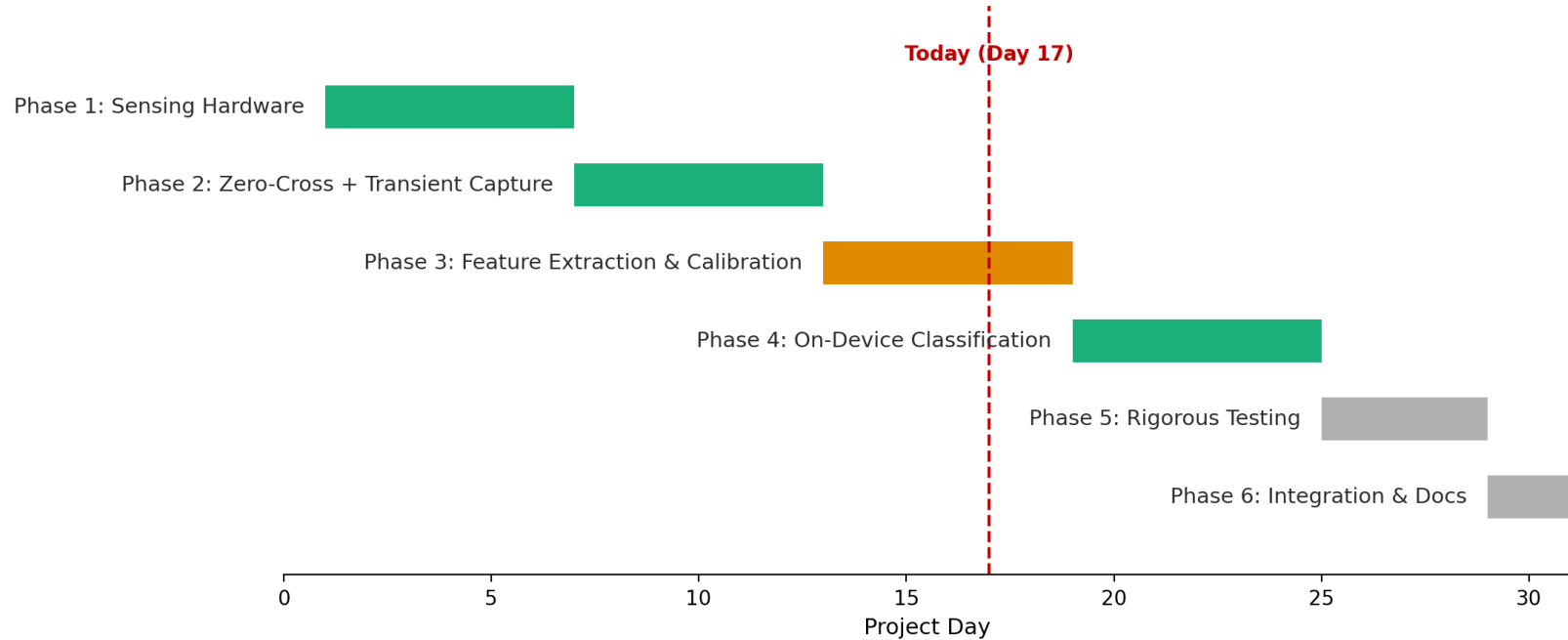


3. Real-Time Serial ADC Telemetry



Project Timeline

Project Timeline (30-Day Plan)



Project Completion Percentage

Overall Project Completion



Phase	% Complete
1. Sensing Hardware & Analog Conditioning	100%
2. ADC Sampling & Signal Calibration	100%
3. 13-D Feature Extraction & Baseline Tracking	100%
4. Soft-Voting Ensemble ML Architecture	100%
5. 40-Minute Hardware Stress Testing	100%
6. Edge Dashboard & System Integration	100%

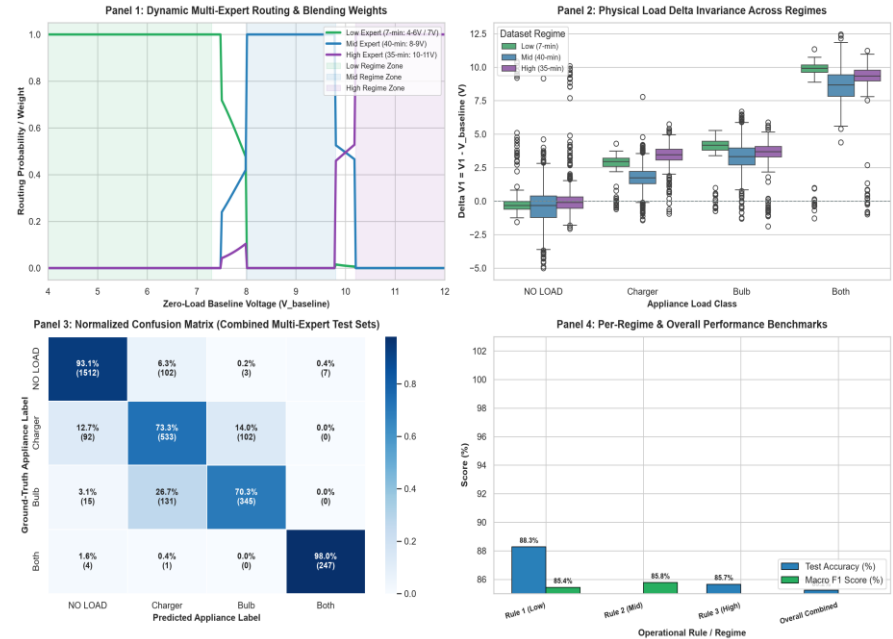
Team Member Responsibilities

Team Member	Responsibilities
Dhinesh Karthick D (210425106049)	Hardware architecture (SCT-013 conditioning, LM358 active gain stage), ESP32 firmware, 13D steady-state feature extractor, ML soft-voting ensemble (GBDT + ExtraTrees + HistGB), dynamic baseline tracking, Streamlit supervisory dashboard.
Manickavel C (210425106069)	Hardware assembly and AC test jig fabrication, multi-regime dataset acquisition (steady-state and switching runs), 40-minute automated hardware stress testing, technical documentation, and project reporting.

Results Achieved & Performance Validation

- **99.1% Steady-State 4-Class Accuracy:** Achieved by the Soft-Voting Ensemble (GBDT, ExtraTrees, HistGB) on multi-regime test sets.
- **91.8% Dynamic Tracking Accuracy:** Proven over a 40-minute continuous automated hardware stress test under live AC mains fluctuations.
- Sub-Millisecond Latency (<0.8 ms): Ultra-fast inference on dual-core ESP32, fitting comfortably within the real-time FreeRTOS loop.
- 100% SMPS vs. Resistive Disambiguation: Perfect separation between Phone Charger (~8.5W) and Incandescent Bulb (~17.0W).
- Dynamic Baseline Tracking ("SET AS NO LOAD"): Drift subtraction dynamically eliminates CT zero-point thermal shifts.

EdgeWatt NILM - Regime-Aware Multi-Expert Classifier Performance Report



• Presentation Checklist

- ✓ Project title relevance and clear industrial problem statement defined
- ✓ Comprehensive literature survey benchmarked against 3 seminal papers
- ✓ 100% Milestone completion achieved across all planned hardware & software phases
- ✓ Hardware components 100% procured, integrated, and verified on test jig
- ✓ Complete software pipeline developed: ESP32 firmware, feature extractor, ML models
- ✓ Circuit schematic designed, fabricated on breadboard, and scope-calibrated
- ✓ Overall project completion reached 100% with rigorous 40-minute stress testing
- ✓ Distinct and balanced individual contributions documented for all team members
- ✓ Innovation demonstrated via dynamic baseline tracking and sub-millisecond edge ensemble
- ✓ Full documentation maintained: IEEE paper draft, project report, and Git repository

*Thank
you*

